

The food chain

In the ocean, nearly all life depends on the food made by seaweeds and the microscopic drifting algae called phytoplankton. They use the energy of sunlight to build living tissue. Tiny animals eat this and turn it into muscle, skin, and other animal tissue. Most of these animals are eaten by other animals, which use the food to build their own bodies. They are eaten by even bigger animals, in a food chain that leads to top predators such as sharks.

EAT AND BE EATEN

Seaweeds and phytoplankton make complex living tissue out of simple chemicals—they are food producers. But animals cannot make their own food; they survive by eating this living (or dead) tissue, so they are food consumers. Animals that eat the algae are primary consumers, while the larger animals that prey on the primary consumers are secondary consumers. These are eaten by even bigger hunters, which fall prey to powerful top predators.



▲ PRODUCER

This seaweed uses solar energy to turn water and carbon dioxide into sugar. It adds other chemicals to turn sugar into living tissue.



▲ PRIMARY CONSUMER

A limpet eats seaweed. In its stomach, the seaweed is digested and used to make substances that help it to grow.



▲ SECONDARY CONSUMER

Crabs cannot digest seaweed. Instead, this shore crab eats limpets that have already turned the seaweed into animal tissue.

FAST FACTS

- The coldest oceans are often the richest in animal life, because cold, stormy waters contain more oxygen and nutrients, which encourage the growth of plankton.
- Animals are scarce in open tropical oceans, because the clear, almost pure water has very little phytoplankton.
- On tropical coral reefs, nearly all the food that supports the reefs is made by tiny algae that live inside the corals.
- In parts of the deep ocean, bacteria make food using energy from volcanic chemicals erupting from the ocean floor.

FOOD PYRAMIDS

This diagram shows that it takes a huge amount of plankton at the bottom of the food chain to support just one top predator, such as this Arctic polar bear. This is because a lot of food is converted into energy and used up before it can be passed up to the next level in the chain.

Zooplankton feed on countless phytoplankton

Phytoplankton make their own food

A polar bear eats dozens of seals per year

Seals eat thousands of fish per year

Fish eat trillions of zooplankton

